

China Data Centers: Following the ROIC

Stronger-than-expected AI infrastructure demand has improved the medium-term growth outlook for China data center operators. However, as order books increasingly convert into long-duration contracted capacity, we believe the sector is becoming less a traditional growth story and more a capital-allocation story. Put differently, we believe **the next stage of China data center investing is less about securing MW capacity or contracts, but more about generating favorable capital returns.**

This distinction has been reinforced by the US rate hikes on Sep 16; GDS/VNET saw a 20%/29% pullback from the recent Sep 21 peak to Oct 2 trough, while the US 10-year Treasury yield has risen to 5.3% currently. Based on our recent conversations, investors are increasingly questioning **whether accelerating AI infrastructure deployment can generate return on invested capital (ROIC) sufficiently above the cost of capital.**

In this report, we **present a ROIC framework** that evaluates and compares the economics of three increasingly capital-intensive data center business models: **1) traditional powered shells, 2) integrated AI data centers (AIDCs) providing GPUaaS, and 3) AI token factories.** Our analysis focuses on the conditions required for each model to earn returns above its cost of capital, and the implications for equity value creation.

Bottom line: ROIC potential rises as operators move closer to AI workload monetization, although execution, technology and earnings risks increase accordingly.

- **Data centers as powered shells remain lower-return assets. Base-case ROIC of c.5%** leaves limited room for value creation through operations alone, and hence **higher equity returns depend on i) financial engineering** i.e. higher financial leverage at lower financing costs e.g. 1.65–1.7% interest rates on Range Intelligent's Rmb3bn onshore debt issued in Aug 2025–Jun 2026, which may increase equity IRR to up to c.10%; **ii) capital recycling** via private and public REITs, which allow cash recycling in sooner years (vs. held to maturity) at a reasonable multiple (i.e. 13–17x EV/EBITDA based on historical transactions) and may increase equity IRR to c.20% on a single project level; and **iii) going global** as higher pricing is likely to more than offset higher construction costs.
- **Integrated AIDCs with long-term GPUaaS contracts:** Revenue and unit capex

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required to produce the same ROIC level vary by GPU IT infrastructure, and we believe **scale operators with superior supply-chain access** are positioned to earn structurally higher returns. To achieve 15% ROIC (same level as assumed by our US TMT team), **the challenge for domestic architectures lies in monetization** as current market rental rates appear insufficient to support a 10-15% ROIC in many cases. However, **for more advanced computing power infrastructure**, the key determinant of ROIC is **securing server supply at the right price**.

- **Integrated AI data centers as token factories:** As a nascent business model, token factories serve inference and application demand primarily on domestic IT infrastructure. Returns become increasingly dependent on **token pricing, model competitiveness, throughput and utilization** rather than contracted infrastructure revenues. As a result, investors can gain significant upside but assume greater earnings volatility and technology risk. That said, we believe **AIDCs carry higher ROIC potential compared to asset-light token factories** that don't own IT infrastructure thanks to vertical integration.

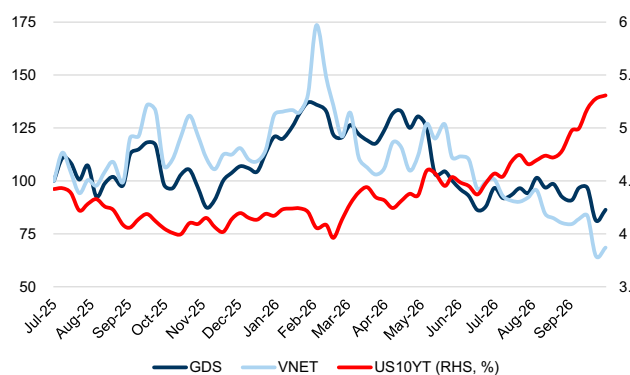
Stock implications:

- **GDS Holdings (GDS/9698.HK, Buy):** Owning powered shells in China and outside China (i.e. c.20% stake in DayOne), we see the expected **C-REIT asset injection** (set to complete in 1H27E, subject to regulatory approval) **as potentially the most important catalyst for GDS China** because it could monetize low-ROIC powered-shell assets and accelerate capital turnover, while the market may not reward order win beats considering limited ROIC appeal against higher US rates (same also applies to VNET in our view).
- **VNET Group (VNET, Buy):** We believe VNET offers a clear pathway to ROIC expansion potential beyond powered shells but execution is key. The **CATL partnership** creates opportunities in compute-energy integration and differentiated competitive advantages, while **overseas projects** (i.e. c.500MW resources secured in Thailand/ Malaysia) and **AIDC potential** (e.g. a zero-carbon token ecosystem) could shift the company's portfolio toward higher-return business models.
- **Range Intelligent (300442.SZ, Buy):** We see Range as best positioned to climb the AI infrastructure value chain as it has executed on three key ROIC advantages: **i) low funding costs and capital recycling** e.g. NDRC's approval for its C-REIT asset injection received in Jun; **ii) AIDC services** including GPUaaS, liquid cooling, etc. has become the core revenue driver, with potentially stepped-up computing power acquisition in 2H26 (see ALC takeaways); **iii) overseas powered shell expansion** starting from HK/Batam towards 1GW delivery in 3-5 years.
- **Athub (603881.SS, Neutral)/Sinnet (300383.SZ, Sell):** While both have initiated GPUaaS investments (e.g. Athub's Rmb1.3bn server expansion plan, Sinnet's Rmb100mn+ annualized contract value on 4k+ PFLOPs computing power), we see their traditional businesses as being under greater growth pressure on limited new capacity (for Athub), and slow customer move-in as well as lower rental pricing (for Sinnet).

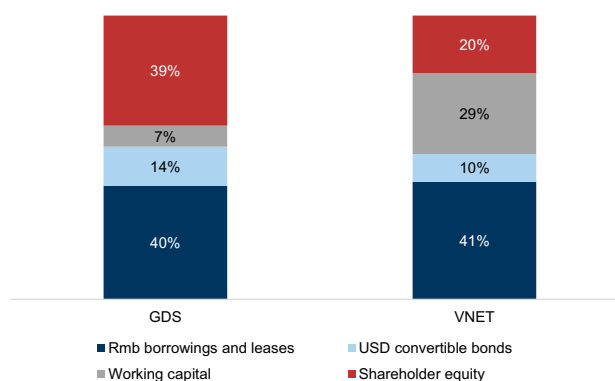
Exhibit 1: Business models, balance sheet and P&L implications of different data center models

	Data Center as Powered Shell	Integrated AI Data Center (GPUaaS)	Integrated AI Data Center (Token Factory)	Asset-Light Token Factory
Business model				
Position in Value Chain	Infrastructure	Compute utility (training oriented)	AI production facility (inference oriented)	Inference/deployment layer
Typical Customers	Hyperscalers, enterprises, AI labs	Hyperscalers, AI labs	Enterprises, vertical AI users	Enterprises, vertical AI users
Typical Contract Duration	7-10 years	3-5 years	Pay-as-you-go (SMEs); monthly to annual subscriptions (KAs)	Pay-as-you-go (SMEs); monthly to annual subscriptions (KAs)
Customer Switching Cost	High (physical migration)	High (data gravity, clusters)	Moderate	Low-Moderate
Economic Moat	- Land, power, permits, construction capex efficiency	- Land, power, permits, construction capex efficiency; - GPU procurement, computing resource orchestration	- Land, power, permits, construction capex efficiency; - GPU procurement, computing resource orchestration; - Distribution, software, workflow integration	- Computing resource supply and orchestration; - Distribution, software, workflow integration
Balance sheet implications				
Capital Intensity	High	Very high	Very high	Low
Working Capital Needs	Low	Low	Low	Moderate
Primary Assets on Balance Sheet	Land, buildings, power infrastructure	Land, buildings, GPUs, networking equipment	Land, buildings, GPUs, networking equipment, software/IP	Right-of-use assets, software/IP
Technology Obsolescence Risk	Low	High	High	Moderate
P&L implications				
Revenue Unit	\$/kW/month	\$/GPU-hour, \$/GPU-month	\$/million tokens, \$/customer	\$/million tokens, \$/customer
Revenue Predictability	Very High	High	Medium	Low-Medium
Revenue Growth Driver	MW deployment and utilization	GPU fleet and utilization	Token throughput, pricing & user growth	Token throughput, pricing & user growth
Utilization Metric	MW utilization	MW x GPU utilization	MW x GPU utilization, token production efficiency	Leased GPU utilization, token production efficiency
Model Price Risk	Low	Moderate	High	High
GPU Price Risk	Low	High	High	Moderate
ROIC Potential	Low	Moderate-High	High	Moderate-High
Key to improve ROIC	1. Faster utilization ramp-up; 2. Access to scarce resources; 3. Higher leverage at lower costs; 4. Capital recycling; 5. Going global	1. Technology infrastructure choice; 2. Disciplined GPU sourcing costs; 3. Premium rental pricing; 4. Hardware obsolescence risk management; 5. Higher leverage at lower costs	1. Technology infrastructure choice; 2. GPU sourcing costs; 3. GPU utilization rate; 4. Token production efficiency; 5. Token pricing; 6. Hardware obsolescence risk management; 7. Higher leverage at lower costs 8. Customer acquisition costs	1. Technology infrastructure choice; 2. GPU sourcing terms; 3. GPU utilization rate; 4. Token production efficiency; 5. Token pricing; 6. Customer acquisition costs

Source: Goldman Sachs Global Investment Research

Exhibit 2: GDS/VNET share prices have corrected -29%/-49% since late Feb when US10YT fell below 4%
 GDS/VNET share prices (Jul 7 = 100) vs. US 10yr Treasury yield


Source: Refinitiv, Data compiled by Goldman Sachs Global Investment Research

Exhibit 3: Breakdown of GDS/VNET assets by funding source (2Q26)

Source: Company data, Goldman Sachs Global Investment Research

1. Data centers as powered shells

Powered shells have been the most prevalent business model among China data centers over the past few years, as operators are responsible for land and building construction, securing the critical utility grid connections and power allotments/equipment, fiber optic line entry pathways, as well as site security and exterior management. By leaving the internal fit-out to tenants, data center operators are much less exposed to rapid technological obsolescence or massive depreciation on short-lived AI hardware on balance sheets, but instead focus on **expanding their footprint and securing resource access (i.e. land, power, and permits)** to meet customer demand and comply with government initiatives.

- **ROIC in our base case:** Our base case assumes Rmb20mn/MW construction capex, Rmb230/kW monthly service revenue (MSR), 95% stabilized utilization, and ~90% EBITDA margin implying **Rmb2.3mn/MW annualized EBITDA**, which are based on the typical contract economics as of late. We estimate **a gross ROIC of c.5%** ([Exhibit 4](#)), which is similar to an unlevered project IRR under a 15-20 years' hold-to-maturity scenario and corresponds to **a levered equity IRR of c.8%** assuming a debt-to-equity ratio of 4x ([Exhibit 5](#)).
- **Key variables that drive ROIC:** MSR and unit capex are the most important drivers ([Exhibit 6](#)), while **utilization rate and ramp-up speed** could affect ROIC and project IRR as delayed customer onboarding or lower rack utilization significantly weakens early-stage returns. However, including **asset monetization** increases returns significantly; assuming Year 7-end exit at 15x EV/EBITDA which is benchmarked against the average multiple implied by historical data center C-REIT/private REIT transactions by GDS, VNET and Range, investment returns could improve to c.15% unlevered IRR and c.24% levered IRR ([Exhibit 7](#)).

Exhibit 4: We estimate a gross ROIC of c.5% for powered shells under our base-case assumptions

Unit economics per 1GW data center in a stabilized stage

Data centers as powered shells (Rmb bn)					
Total capex	20.0	A = a+b+c	Revenue	2.6	C = e*f*12
Land and others	0.3	a	MSR (Rmb/W)	0.23	e
Land area (mu)	533		Utilization	95%	f
Land cost (Rmb k/mu)	500		Owner costs	0.3	D
Facility	4.8	b	Labor and others	0.3	
GFA (k sqm)	1,067		Maintenance headcount	1,000	
Cost per GFA (Rmb k/sqm)	4.5		Monthly cost (Rmb k)	23	
Electrical infrastructure	15	c	EBITDA	2.3	E = C-D
Depreciation cost p.a.	1.2	B = A/d	EBITDA margin	89%	
Depreciation years	17	d	Depreciation cost	1.2	B
Land	40		EBIT	1.2	F=E-B
Building construction	30		Tax rate	20%	g
Electrical equipment	15		NOPAT	0.9	G=F*(1-g)
Gross ROIC	5%	H = G/A			
Lifecycle return %	81%	I = H*d			

Source: Goldman Sachs Global Investment Research

Exhibit 5: We estimate an unlevered IRR of 5% for powered shells under a hold-to-maturity scenario

Unit economics per 1GW data center under a hold-to-maturity scenario

Rmb bn per 1GW / year	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Key assumptions	
Utilization - period end	0%	0%	50%	90%	95%	95%	95%	95%	95%	95%	95%	95%	95%	95%	95%	95%	95%	95%	Construction Capex	20
Utilization - period avg	0%	0%	25%	70%	93%	95%	95%	95%	95%	95%	95%	95%	95%	95%	95%	95%	95%	95%	Upgrade Capex (every 10yrs)	0.25
Revenue (Rmb/kw)	-	-	0.7	1.9	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	Depreciation years	17
EBITDA	-	-	0.5	1.7	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3	MSR (Rmb/kW)	230
EBITDA margin			73%	87%	89%	89%	89%	89%	89%	89%	89%	89%	89%	89%	88%	88%	88%	88%	Cost inflation	1%
Capex	(10.0)	(10.0)									(0.3)								EBITDA margin	89%
Depreciation		0.6	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	Utilization	95%
as % revenue			168%	60%	46%	44%	44%	44%	44%	44%	44%	45%	45%	45%	45%	45%	45%	45%		
EBIT	(0.6)	(0.7)	0.5	1.1	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.1	1.1	1.1	1.1	1.1	1.1	1.1	Debt/Asset	80%
COGS ex. depreciation	-	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	Cost of debt	3.9%
as % revenue			27%	13%	11%	11%	11%	11%	11%	11%	11%	11%	11%	11%	12%	12%	12%	12%	Debt tenure (years)	10
Tax	-	-	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	Cost of equity	12%
ROIC			2%	5%	6%	7%	7%	8%	9%	10%	11%	12%	15%	18%	23%	33%	58%		Tax rate	20%
FCF unlevered	(10.0)	(10.0)	0.5	1.6	2.1	2.1	2.1	2.1	2.1	2.1	1.8	2.1	2.1	2.1	2.1	2.1	2.1	2.1		
Interest expenses			(1)	(1)	(1)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)						Returns	
Tax shield on interest			0	0	0	0	0	0	0	0	0	0	0						Unlevered IRR	5%
Debt repayment			(1)	(1)	(1)	(2)	(2)	(2)	(2)	(2)	(2)	(2)							Levered IRR	8%
FCF levered	(2.0)	(2.0)	(1.3)	(0.3)	0.2	0.2	0.2	0.2	0.2	0.2	(0.1)	0.2	2.1	2.1	2.1	2.1	2.1	2.1		

Source: Goldman Sachs Global Investment Research

Exhibit 6: Powered shell ROIC sensitivity

Powered-shell ROIC sensitivity		Total capex (Rmb bn)				
		25.0	22.5	20.0	17.5	15.0
MSR (Rmb/W)	0.17	1%	2%	2%	4%	5%
	0.20	2%	3%	4%	5%	7%
	0.23	3%	4%	5%	7%	9%
	0.26	4%	5%	7%	8%	11%
	0.29	5%	7%	8%	10%	13%

Source: Goldman Sachs Global Investment Research

Exhibit 7: Asset monetization can increase powered shell IRR vs. hold-to-maturity

Unit economics per 1GW data center under an asset monetization scenario

Rmb bn per 1GW / year	0	1	2	3	4	5	6	7
Exit EV/EBITDA								15.0x
EV								35.1
FCF unlevered	(10.0)	(10.0)	0.5	1.6	2.1	2.1	2.1	37.2
Unlevered IRR								15%
EV								35.1
Net debt								(16.0)
Net proceeds								19.1
FCF levered	(2.0)	(2.0)	(1.3)	(0.3)	0.2	0.2	0.2	19.4
Levered IRR								24%

Exit EV / EBITDA	10.0x	12.0x	14.0x	16.0x	18.0x
Unlevered IRR	9%	12%	14%	16%	17%
Levered IRR	7%	15%	21%	26%	30%

Source: Goldman Sachs Global Investment Research

2. Integrated AI data centers with long-term GPUaaS contracts

The robust demand for computing power from hyperscalers and AI labs, coupled with a favorable onshore funding environment, is enabling a number of data center operators (e.g. Range Intelligent, Guangdong HEC) to transform their business models from traditional powered shells to **more capital intensive AI data centers** that provide customers with **advanced computing power access** under 3-5 years' contracts in addition to powered shells. In some cases, operators provide **turnkey solutions** to customers that include deeply tuned networking capabilities, AI-focused software, liquid cooling, etc.

Instead of estimating ROIC based on a number of base-case assumptions as in the case of powered shells, our analysis in this report focuses on the **revenue and unit capex hurdles required to produce a 15% annual ROIC for a 1GW deployment** which is similar to the methodology applied in Americas Technology: Sizing the AI Economy Needed to Justify ROIC on Hyperscaler AI Capex. This considers the various GPU IT infrastructure (Exhibit 8) of this emerging business model, and different cost/pricing levels of operators, as opposed to relatively standardized economics for powered shells.

Exhibit 8: Specs and economics of a 200MW IT power data center based on different IT infrastructure

Capex per 200MW IT power data center	Nvidia B300	Nvidia H800	Huawei CM384 910C	Huawei CM1024 950	vs. 910C	vs. B300
Computing power (EFLOPS, FP16)	250	240	80	76		
Computing power (EFLOPS, FP8)	500	480	80	142		
Total IT power (kW)	187,500	190,400	208,800	199,066		
Server/supernode units #	12,500	20,000	261	135		
GPUs # per unit	8	8	384	1,024		
CPUs # per unit	2	2	192	256		
Computing power per unit (PFLOPS, FP16)	20	12	307	563		
Computing power per GPU (TFLOPS, FP16)	2,500	1,500	800	550		
Computing power per GPU (TFLOPS, FP8)	5,000	3,000	800	1,024		
IT power per unit (kW)	15	10	800	1,475		
TDP per GPU (W)	1,400	700	800	600		
Coefficient	1.3	1.7	2.6	2.4		
IT power per server (kW)	15	10	17	12		
Total capex (Rmb bn)	73	68	41	58		
Capex per IT power (Rmb bn/GW)	392	355	196	291	1.5x	0.7x
Capex per computing power (Rmb bn/EFLOPS, FP8)	0.15	0.14	0.51	0.41	0.8x	2.8x
Computing power per IT power (EFLOPS/GW, FP8)	2,667	2,521	384	711	1.9x	0.3x

Source: Company data, Goldman Sachs Global Investment Research

- **Revenue and unit capex hurdles to meet 15% ROIC** vary by technology selection (e.g. Nvidia vs. domestic architectures), given meaningful differences in capex intensity, compute output, and monetization potential. To achieve 15% ROIC, we see **unit capex as the key hurdle for more advanced computing power, while rental revenue is the key hurdle for domestic servers.**

For example, leasing out Nvidia B300-equivalent servers at a monthly rate of Rmb192k/server could generate a 15% ROIC, against the upfront capex at Rmb5.5mn per server; based on our company checks, the Rmb192k/server monthly rental rate appears to be well below market rates (though long-term contracts may provide price discounts) while Rmb5.5mn unit capex per B300-equivalent server would be less achievable for smaller operators considering price hikes YTD towards as high as Rmb15mn. However, rental revenue hurdles for domestic servers to meet 15% ROIC appear to be too high vs. actual market rates ([Exhibit 11](#)) implying lower actual ROIC in our view.

- **Key variables that drive ROIC:** Unlike the powered shell model where value creation is primarily driven by asset monetization and leverage, achieving attractive equity returns of AIDC depends on **disciplined GPU procurement costs, premium rental pricing, and hardware obsolescence risk management** against a very large upfront capital commitment. **Timing** of computing power procurement and leasing out could also affect ROIC as some companies even re-negotiate rental prices with customers post initial contract signs, in the context of surging GPU rental prices based on market rates ([Exhibit 11](#)).
- **Funding as a key variable to equity returns:** To fund these capital intensive investments, secured bank loans and/or finance leasing have become the go-to channels with funding costs at LSD-MSD % and debt-to-equity ratios at 4-10x, based on our checks. In other words, we believe **AIDCs with long-term GPUaaS contract can enjoy more flexible low-cost financing than powered shells**, potentially leading to higher equity returns at the same ROIC.

Exhibit 9: Unit economics per 1GW AIDC with LT GPUaaS contracts assuming 15% annual ROIC

Integrated AIDC with LT GPUaaS contracts (Rmb bn)	Nvidia B300	Nvidia H800	Huawei CM384 910C	Huawei CM1024 950	
Total capex (Rmb bn)	392	355	196	291	A = a+b
Servers/supernodes	367	330	169	244	a = a1*a3
Server/supernode units #	66,667	105,042	1,250	678	a1
GPUs # per unit	8	8	384	1,024	a2
Unit cost (Rmb mn)	5.5	3.1	135.0	360.5	a3
Land, facility, electrical infra and others	25	25	28	47	b
Depreciation cost p.a.	75	67	35	51	B = A/c
Depreciation years	5.2	5.3	5.5	5.7	c
Revenue	154	140	78	112	C = d+E
Revenue per GW (US\$ bn)	23	21	11	17	C1 = C/6.8
Monthly revenue per server (Rmb k)	192	111	108	108	C1 = C/a1/(a2/8)/12*10^6
Owner costs	5.6	5.6	5.6	6.5	d = e+f
Electricity cost	4.7	4.7	4.7	4.7	e = e1*e2*365*24/10^3
Electricity cost (Rmb/kWh)	0.45	0.45	0.45	0.45	e1
PUE	1.2	1.2	1.2	1.2	e2
Labor and others	0.9	0.9	0.9	1.8	f
EBITDA	148	134	72	106	E = B+F
Depreciation cost	75	67	35	51	B
EBIT	73	67	37	55	F = G/(1-g)
EBIT margin	48%	48%	47%	49%	g
Tax rate	20%	20%	20%	20%	G = A/H
NOPAT	59	53	29	44	
NOPAT margin	38%	38%	38%	39%	
Gross ROIC	15%	15%	15%	15%	H

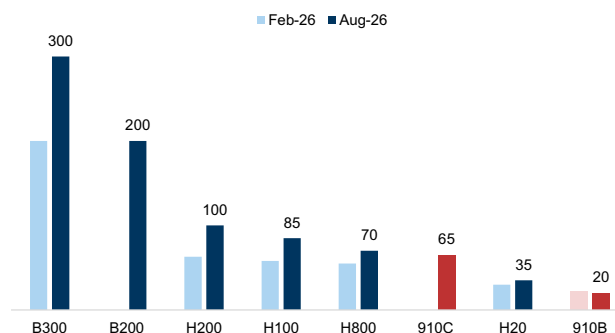
Source: Company data, Goldman Sachs Global Investment Research

Exhibit 10: Monthly revenue per server sensitivity for Nvidia B300 and Huawei 910C (Rmb k)

		Unit cost per Nvidia B300 server (Rmb mn)				
		5.5	7.5	9.5	11.5	13.5
ROIC	0%	101	134	167	201	234
	5%	131	175	219	262	306
	10%	162	216	270	324	378
	15%	192	257	322	386	451
	20%	223	298	373	448	523

		Unit cost per Huawei CM384 910C supernode (Rmb mn)				
		75	95	115	135	155
ROIC	0%	36	43	50	57	64
	5%	47	56	65	74	83
	10%	57	69	80	91	102
	15%	68	81	95	108	122
	20%	78	94	110	125	141

Source: Goldman Sachs Global Investment Research

Exhibit 11: Monthly rental fee per server for different IT infrastructure based on market rates (Rmb k)


Source: Company data, Data compiled by Goldman Sachs Global Investment Research

3. Integrated AI data centers as token factories

Unlike AI data centers with long-term GPUaaS contracts, which typically generate predictable cash flows and stable ROICs but capture only a limited share of the value created by downstream AI applications, the token factory model is focused on monetizing compute capacity through AI workloads, converting owned or leased computing resources into revenue-generating token output.

- **Asset-light token factory:** In our framework for an asset-light token factory ([Exhibit 12](#)), we assume a 1GW deployment supported by long-term GPU lease contracts, where the operator incurs fixed annual compute lease costs while revenues are generated from inference token production.

Under the B300-equivalent scenario, our assumptions imply **i) Inference service revenue:** annual token revenue of c.Rmb288bn, supported by assumed token price (US\$0.69/mn tokens based on DeepSeek V4 Pro as a reference [Exhibit 19](#)), token

throughput, utilization assumptions. **ii) GPU lease costs** based on a monthly GPU lease rate of Rmb192k/server or c.Rmb154bn annually for 1GW of compute (consistent with the rental revenue assumptions used in the GPU rental business model from the perspective of a long-term GPUaaS provider). **iii) NOPAT calculations:** Future GPUaaS lease payments are capitalized into a right-of-use asset with annual depreciation, and this translates to EBIT of Rmb162bn or NOPAT of c.Rmb129bn per year. **iv) Invested capital and ROIC:** Against a gross right-of-use asset of c.Rmb631bn, representing the present value of 5-year GPU leasing commitments, the implied annual ROIC is approximately 21%.

Exhibit 12: Unit economics per 1GW IT power secured by an asset-light AI token factory

Asset-light AI token factory	Nvidia B300	Nvidia H800	Huawei CM384 910C	Huawei CM1024 950	
Inference service revenue (Rmb bn)	288	151	58	144	A = a*b
Token price (Rmb/mn)	4.7	4.7	4.7	4.7	a = a1*6.8
Selected token price (US\$/mn)	0.69	0.69	0.69	0.69	a1 <-- blended price for DeepSeek V4 Pro
Tokens per GW-year (quadrillion)	61	32	12	31	b = b1*b2*b3*b4*b5 *365*24*3600/10^12
Share of compute on inference	100%	100%	100%	100%	b1
Inference utilization rate	70%	70%	70%	70%	b2
Production efficiency factor	1.0x	1.0x	1.0x	1.0x	b3
GPU token throughput/sec	5,208	1,736	1,157	2,000	b4 <-- input from IT specs
GPUs # per GW (k)	533	840	480	694	b5 <-- input from IT specs
Incremental service cost (Rmb bn)	154	140	78	112	B = c1+A*c2
GPUs rental cost (Rmb bn)	154	140	78	112	c1 <-- Input from C in Ex.9
Electricity cost (included in AIDC rent)	-	-	-	-	c2
Revenue split with LLMs (% inference revenue)	0%	0%	0%	0%	C = A-B
Service-layer incremental profit (Rmb bn)	134	12	(20)	32	D = C/A
Service-layer incremental margin	47%	8%	(35%)	22%	E = c1/(1-c2)/b/6.8
Breakeven token price (US\$/mn)	0.37	0.64	0.93	0.54	
Initial ROU asset/ lease liability (Rmb bn)	631	572	319	461	F = PV(e, d, c1*-1)
Lease term (years)	5	5	5	5	d
Discount rate	7%	7%	7%	7%	e
Lease-adjusted EBIT (Rmb bn)	162	37	(6)	52	G = A-f-g
Annual ROU depreciation (Rmb bn)	126	114	64	92	f = F/d
Revenue split cost (Rmb bn)	-	-	-	-	g
Lease-adjusted NOPAT (Rmb bn)	129	29	(6)	42	H = G*(1-i)
Tax rate	20%	20%	20%	20%	i
Gross lease ROIC p.a.	21%	5%	-2%	9%	I = H/F

Source: CAICT, Goldman Sachs Global Investment Research

Exhibit 13: Sensitivity of token price (US\$/mn tokens) on ROIC under an asset-light token factory model, else equal

		Nvidia B300	Nvidia H800	Huawei CM384 910C	Huawei CM1024 950
Token price (US\$/mn)	0.10	-13%	-16%	-17%	-15%
	0.30	0%	-9%	-12%	-6%
	0.69	21%	5%	-2%	9%
	1.20	47%	21%	9%	27%
	2.50	116%	61%	36%	75%

Source: Goldman Sachs Global Investment Research

Exhibit 14: Sensitivity of inference utilization rate on ROIC under an asset-light token factory model (using US\$0.69/mn token price)

		Nvidia B300	Nvidia H800	Huawei CM384 910C	Huawei CM1024 950
Inference utilization rate	50%	10%	-1%	-7%	2%
	60%	15%	2%	-5%	5%
	70%	21%	5%	-2%	9%
	80%	26%	8%	0%	13%
	90%	31%	11%	3%	16%

Source: Goldman Sachs Global Investment Research

Exhibit 15: Sensitivity of monthly server rental costs on ROIC under an asset-light token factory model (using US\$0.69/mn token price)

		Nvidia B300		Nvidia H800		Huawei CM384		Huawei CM1024			
Monthly server rental costs (Rmb k/server)	130	38%	Monthly server rental costs (Rmb k/server)	71	17%	Monthly server rental costs (Rmb k/server)	39	24%	Monthly server rental costs (Rmb k/server)	60	29%
	161	28%		91	10%		73	5%		84	16%
	192	21%		111	5%		108	-2%		108	9%
	224	15%		131	2%		143	-6%		132	4%
	255	12%		150	-1%		178	-9%		156	1%

Source: Goldman Sachs Global Investment Research

- **Token factory with owned GPUs:** Under the Nvidia B300-equivalent scenario, we

continue to assume inference service revenue to be Rmb288bn. After deducting electricity, labor and depreciation expenses, this translates into Rmb209bn of EBIT and Rmb167bn of NOPAT. Against c.Rmb392bn of gross allocated capex, the implied ROIC reaches c.43%. This scenario illustrates the return upside available under the token factory model when operators directly own the underlying GPU infrastructure.

- **Key variables that drive ROIC:** The token factory model introduces greater earnings volatility vs. a GPU rental model. While GPU procurement/rental costs and financing structure remain important, the primary drivers of token factory economics are token pricing, GPU token throughput, compute density and inference utilization rates. Among these variables, GPU token throughput is particularly important. The superior throughput Nvidia B300-equivalent servers translates into substantially higher capital productivity and ROIC, whereas alternative architectures generally require higher utilization levels and/or stronger token monetization to generate comparable returns.

Exhibit 16: Unit economics per 1GW AIDC as a token factory

Integrated AIDC as a token factory	Nvidia B300	Nvidia H800	Huawei CM384 910C	Huawei CM1024 950	
Inference revenue (Rmb bn)	288	151	58	144	A = a*b
Token price (Rmb/mn)	4.7	4.7	4.7	4.7	a = Reference a in Ex. 12
Selected token price (US\$/mn)	0.69	0.69	0.69	0.69	Reference a1 in Ex. 12
Tokens per GW-year (quadrillion)	61	32	12	31	b = Reference b in Ex. 12
Share of compute on inference	100%	100%	100%	100%	b1 = Reference b1 in Ex. 12
Inference utilization rate	70%	70%	70%	70%	b2 = Reference b2 in Ex. 12
Production efficiency factor	1.0x	1.0x	1.0x	1.0x	Reference b3 in Ex. 12
GPU token throughput/sec	5,208	1,736	1,157	2,000	Reference b4 in Ex. 12
GPUs # per GW (k)	533	840	480	694	Reference b5 in Ex. 12
Costs and expenses (Rmb bn)	79	72	40	56	B = c1+c2+c3
Electricity cost	3	3	3	3	c1 = Reference e in Ex. 9*b1*b2
Labor and other cost	1	1	1	2	c2 = Reference f in Ex. 9*b1
Depreciation	75	67	35	51	c3 = Reference B in Ex. 9*b1
EBIT (Rmb bn)	209	80	18	88	C = A-B
Tax rate	20%	20%	20%	20%	d
NOPAT (Rmb bn)	167	64	14	70	D = C*(1-d)
Allocated capex — gross (Rmb bn)	392	355	196	291	E = Reference A in Ex. 9*b1
Gross ROIC	43%	18%	7%	24%	F = D/E

Source: Goldman Sachs Global Investment Research

Exhibit 17: Sensitivity of token price (US\$/mn tokens) on ROIC under a token factory model with owned GPUs, else equal

		Nvidia B300	Nvidia H800	Huawei CM384 910C	Huawei CM1024 950
Token price (US\$/mn)	0.10	-10%	-14%	-16%	-12%
	0.30	9%	-2%	-7%	2%
	0.69	43%	18%	7%	24%
	1.20	86%	43%	25%	53%
	2.50	197%	107%	69%	128%

Source: Goldman Sachs Global Investment Research

Exhibit 18: Sensitivity of inference utilization rate on ROIC under a token factory model with owned GPUs

		Nvidia B300	Nvidia H800	Huawei CM384 910C	Huawei CM1024 950
Inference utilization rate	50%	26%	8%	1%	13%
	60%	34%	13%	4%	19%
	70%	43%	18%	7%	24%
	80%	51%	23%	10%	30%
	90%	59%	27%	14%	35%

Source: Goldman Sachs Global Investment Research

Exhibit 19: Model price vs. Artificial Analysis Intelligence Index and Arena.ai scores

Price (US\$ per mn tokens)	Input (cache hit)	Input (cache miss)	Output	Blended	Artificial Analysis Intelligence Index	Agent Arena Net Improvement %	Code Arena WebDev Score	Text Arena Score
MiMo-V2.6-Flash	0.003	0.14	0.28	0.06	38	-0.6%	1643	1452
GPT-6 Luna	0.010	0.10	0.50	0.08	38	1.4%	1579	1443
GLM-5.3-Flash	0.026	0.15	0.50	0.10	42	-0.4%	1616	1473
Hy3	0.034	0.14	0.56	0.11	27	-5.4%	1509	1456
MiMo-V2.6-Pro	0.004	0.44	0.87	0.18	46	3.3%	n.a	1480
DeepSeek V4.1 Flash	0.006	0.30	1.20	0.18	39	4.0%	1620	1474
MiniMax-M3	0.060	0.30	1.20	0.22	29	-6.9%	1482	1440
Qwen3.8 27B	0.050	0.50	3.00	0.44	34	-1.7%	1591	1438
DeepSeek V4 Pro	0.044	1.32	3.96	0.69	41	1.2%	1583	1458
GLM-5.3 Max	0.260	1.40	4.40	0.90	45	2.4%	1623	1478
Qwen3.8 Max	0.250	2.00	6.00	1.18	46	2.5%	1671	1482
Grok 4.7	0.500	2.00	6.00	1.35	52	4.0%	1638	1442
GPT-6.1 Sol	0.100	2.00	10.00	1.47	54	11.2%	1758	1483
Gemini 4 Argon	0.100	2.00	10.00	1.47	54	7.6%	1680	1525
MiMo-V2.6-Pro-UltraSpeed	0.036	4.35	8.70	1.77	46	3.3%	n.a	1480
Kimi K3	0.300	3.00	15.00	2.31	47	4.2%	1658	1488
Claude Opus 5.5	0.200	4.00	20.00	2.94	61	13.8%	1815	1504
Claude Fable 5.1	0.250	10.00	50.00	7.18	56	14.3%	1749	1501
GPT-6 Astra	1.000	10.00	50.00	7.70	55	12.3%	1788	1477

Blended price is based on the 7:2:1 mix of cache-hit input, cache-miss input, and output.

Source: Company data, Artificial Analysis, arena.ai, Data compiled by Goldman Sachs Global Investment Research

Price Target Risks and Methodology - GDS Holdings

Our 12m target prices of **US\$49/HK\$48** for GDS/9698.HK are based on a SOTP valuation for GDS China and DayOne, with a 10% holdco discount. Key downside risks: Below-expected move-in demand and utilization improvement, slower overseas revenue/profitability ramp-up, lower-than-expected pricing trend in China and overseas markets, customer churn, slower deleveraging process.

Price Target Risks and Methodology - VNET Group

We have a 12m target price of US\$14, based on a target 12m-fwd EV/EBITDA multiple of 10.5x applied to 2027E adj. EBITDA. Key risks: 1) inability to finance the growth objectives; 2) softer-than-expected execution on order wins; 3) geopolitical risks regarding AI; 4) further downturn of traditional businesses; 5) faster than or unexpected change in AI model training demand brought by the latest development in technology.

Price Target Risks and Methodology - Range Intelligent

Valuation: Our 12-month target price of Rmb109 is based on an 20x 2030E EV/EBITDA discounted back to end-2026E at a 10% CoE. The 20x target EV/EBITDA multiple is a weighted average of: i) a 25x target EV/EBITDA multiple to the DC operation EBITDA and ii) a 10x target EV/EBITDA multiple to GPUaaS.

Key risks: 1) Lower-than-expected order wins amid a competitive environment; 2) Slower-than-expected utilization rate ramp-up; 3) Larger-than-expected pricing pressure; 4) Worse-than-expected execution in overseas expansion; 5) Changes in chip availability due to regulatory changes, domestic capacity ramp-up delays, etc; 6) Difficulties in financing.

Price Target Risks and Methodology - Shanghai Athub

We are **Neutral** rated on Shanghai Athub with a 12m target price of Rmb27.5, based on a 20x target 12m-fwd EV/EBITDA applied to 2027E EBITDA. Key risks include: 1) Above/below-expected delivery/move-in of the Langfang AIDC project; 2) High

customer concentration risks and benefits; 3) Slower-than-expected execution on capacity expansion; 4) Below-expected execution on AIDC expansion; 5) Contract renewal uncertainty.

Price Target Risks and Methodology - Beijing Sinnet Technology Co Ltd.

We are Sell rated on Sinnet Technology with a 12m target price of Rmb11.2, based on a 16x target 12m-fwd EV/EBITDA. Key upside risks: 1) Stronger IDC demand recovery and faster capacity delivery/move-ins, 2) key order wins, 3) lower-than-expected power costs and stricter SG&A expense control, 4) stronger execution on new initiatives including AIDC and overseas expansion.

Disclosure Appendix

Reg AC

I, Timothy Zhao, hereby certify that all of the views expressed in this report accurately reflect my personal views about the subject company or companies and its or their securities. I also certify that no part of my compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

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Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

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Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DAFCF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

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Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

Rating and pricing information

Beijing Sinnet Technology Co Ltd. (Sell, Rmb10.27), GDS Holdings (ADR) (Buy, \$31.63), GDS Holdings (H) (Buy, HK\$29.76), Range Intelligent (Buy, Rmb62.43), Shanghai Athub (Neutral, Rmb22.35) and VNET Group (Buy, \$5.39)

The rating(s) for Beijing Sinnet Technology Co Ltd., GDS Holdings (ADR), GDS Holdings (H), Range Intelligent, Shanghai Athub and VNET Group is/are relative to the other companies in its/their coverage universe: Autohome Inc. (ADR), Autohome Inc. (H), Beijing Sinnet Technology Co Ltd., East Buy, GDS Holdings (ADR), GDS Holdings (H), KE Holdings (ADR), KE Holdings (H), Kanzhun Ltd. (ADR), Kanzhun Ltd. (H), Kingsoft Cloud, New Oriental Education & Technology (ADR), New Oriental Education & Technology (H), Offcn Education Technology, Range Intelligent, SUNeVision Holdings, Shanghai Athub, TAL Education Group, Tuohu Car Inc., Tuya, VNET Group, Weibo Corp. (ADR), Weibo Corp. (H), Weimob, Xiaomi Corp.

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Goldman Sachs had a non-securities services client relationship during the past 12 months with: VNET Group (\$5.17)

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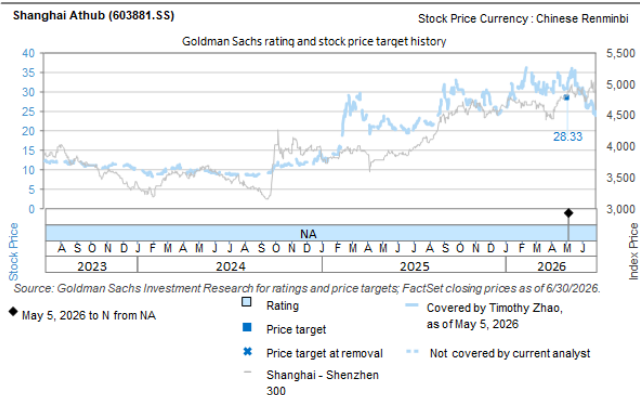
Distribution of ratings/investment banking relationships

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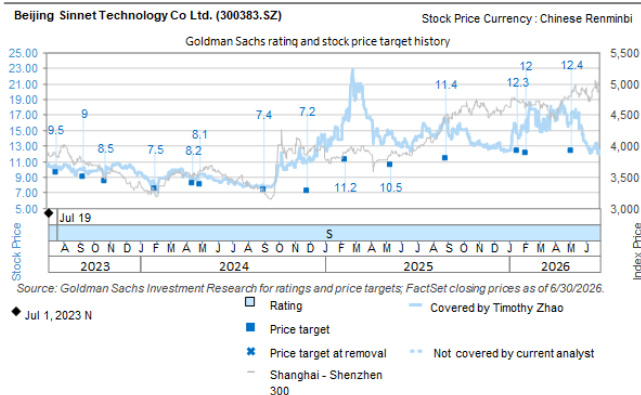
	Rating Distribution				Investment Banking Relationships		
	Buy	Hold	Sell		Buy	Hold	Sell
Global	50%	34%	16%		65%	59%	43%

As of July 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,104 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

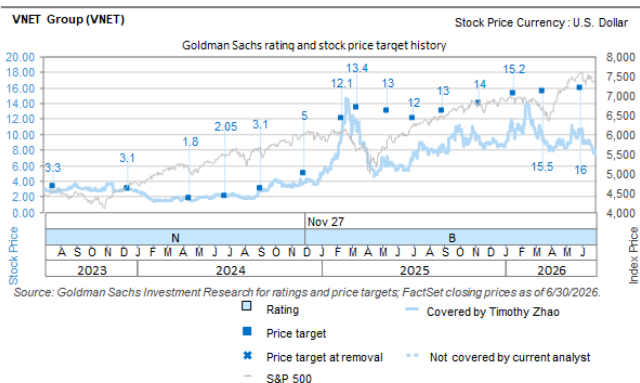
Price target and rating history chart(s)



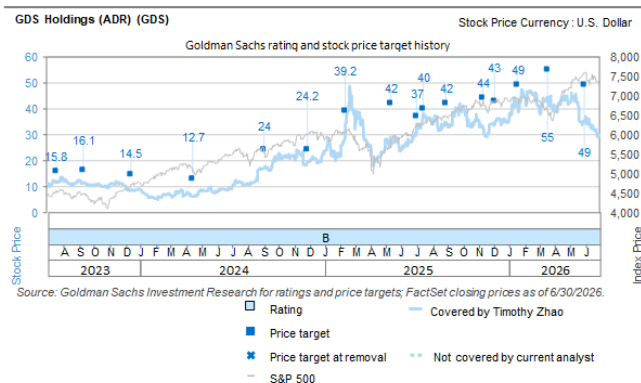
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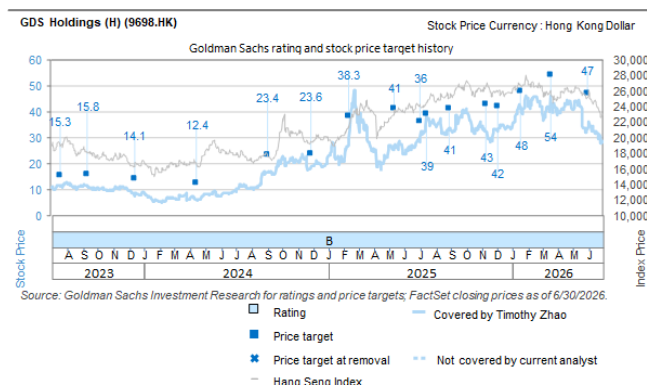
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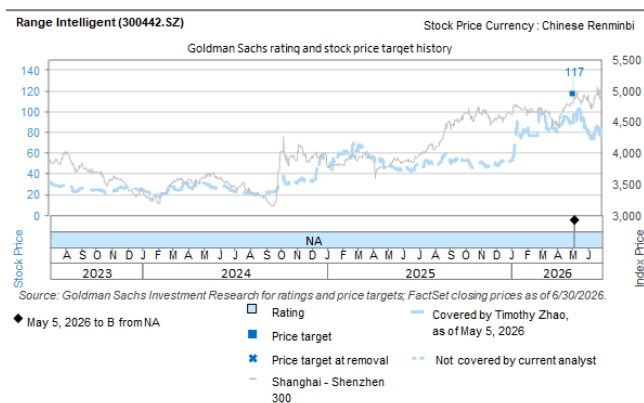
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The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.

Target price history table(s)

GDS Holdings (H) (9698.HK)

Date of report	Target price (HK\$)	Closing price (HK\$)
16-Aug-26	48.00	34.24
27-Jul-26	45.00	31.16
30-May-26	47.00	33.28
18-Mar-26	54.00	45.98
18-Jan-26	48.00	41.86
03-Dec-25	42.00	32.50
11-Nov-25	43.00	32.60
29-Aug-25	41.00	33.76
15-Jul-25	39.00	38.65
04-Jul-25	36.00	32.25
12-May-25	41.00	27.20
11-Feb-25	38.30	30.00
27-Nov-24	23.60	18.28
03-Sep-24	23.40	16.70
15-Apr-24	12.40	6.17
14-Dec-23	14.10	8.34

VNET Group (VNET)

Date of report	Target price (\$)	Closing price (\$)
23-Aug-26	14.00	6.72
30-May-26	16.00	10.08
16-Mar-26	15.50	9.53
18-Jan-26	15.20	10.56
11-Nov-25	14.00	9.79
29-Aug-25	13.00	8.66
04-Jul-25	12.00	7.75
12-May-25	13.00	7.23
12-Mar-25	13.40	11.00
11-Feb-25	12.10	9.97
27-Nov-24	5.00	3.86
03-Sep-24	3.10	2.71
25-Jun-24	2.05	2.24
15-Apr-24	1.80	1.58
14-Dec-23	3.10	2.94

Beijing Sinnet Technology Co Ltd. (300383.SZ)

Date of report	Target price (Rmb)	Closing price (Rmb)
04-Sep-26	11.20	12.17
06-May-26	12.40	17.29
03-Feb-26	12.00	16.40
18-Jan-26	12.30	14.73
29-Aug-25	11.40	16.15
12-May-25	10.50	14.76
11-Feb-25	11.20	16.34
27-Nov-24	7.20	11.13
03-Sep-24	7.40	7.92
29-Apr-24	8.10	9.41
15-Apr-24	8.20	8.90
31-Jan-24	7.50	7.78
24-Oct-23	8.50	9.30

Shanghai Athub (603881.SS)

Date of report	Target price (Rmb)	Closing price (Rmb)
04-Sep-26	27.50	24.72
05-May-26	34.00	30.76

GDS Holdings (ADR) (GDS)

Date of report	Target price (\$)	Closing price (\$)
16-Aug-26	49.00	34.41
27-Jul-26	46.00	31.37
30-May-26	49.00	35.45
18-Mar-26	55.00	44.49
18-Jan-26	49.00	40.66

Range Intelligent (300442.SZ)

Date of report	Target price (Rmb)	Closing price (Rmb)
04-Sep-26	109.00	62.93
05-May-26	117.00	89.06

GDS Holdings (ADR) (GDS)

Date of report	Target price (\$)	Closing price (\$)
03-Dec-25	43.00	33.74
11-Nov-25	44.00	33.33
29-Aug-25	42.00	34.56
15-Jul-25	40.00	37.72
04-Jul-25	37.00	33.98
12-May-25	42.00	29.13
11-Feb-25	39.20	34.29
27-Nov-24	24.20	19.54
03-Sep-24	24.00	16.98
15-Apr-24	12.70	6.12
14-Dec-23	14.50	8.78

Range Intelligent (300442.SZ)

Date of report	Target price (Rmb)	Closing price (Rmb)
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Price targets shown in table(s) are unadjusted for corporate actions.

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